

MAIB AI 208 · AI IN MARKETING · PROJECT REPORT

RetentionLens

Ninety-day lapse risk and customer value for every wholesale account, the evidence behind each call, and outreach a person approves before anything leaves.

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Live demo: krish2105.github.io/Retention-Lens-AI-in-Marketing-

Code: github.com/krish2105/Retention-Lens-AI-in-Marketing-

What was built and what it shows

RetentionLens scores 5,852 trade accounts from a real UK giftware wholesaler (UCI Online Retail II, 1,067,371 invoice lines, CC BY 4.0) for the chance they place no order in the next 90 days and for their expected spend. It explains each call, groups accounts into segments, adds web and network signals, and drafts a check-in email with a local language model that a person must approve. Since the first version it also accepts a company's own invoice export, builds a weekly review queue with a random holdout to measure whether outreach works, and has the foundations of a multi-tenant service (workspaces, roles, row-level security, load and restore tests). None of this has yet been used by a real customer.

3,124	£467k	0.820	0.821
accounts at high lapse risk (≥ 75%)	usual 90-day spend at stake in them	PR-AUC of BG/NBD on the frozen test	AUC of the new-customer lookalike model

Main findings

- **Simple probabilistic models win.** Across four rolling cutoffs, LightGBM beat BG/NBD on 0 and logistic RFM on 1. A neural model is not justified by this evidence.
- **A seasonal correction halves the spend bias.** Mean total-spend bias on the development cutoffs fell from 32.9% to 16.8%; a ZILN neural challenger won 0 of 3 cutoffs on lapse PR-AUC and was not adopted.
- **Wholesale buying leads consumer interest by about a month.** For Christmas lines the correlation with Wikipedia interest rises from 0.34 (same month) to 0.90 (one month later). Twelve monthly points: exploratory.
- **New customers can be spotted from their first order** (test AUC 0.821 on later customers).
- **A small local LLM needs a person in the loop.** 15/16 drafts kept to the model's numbers; its own second-pass review passed only 1/16.

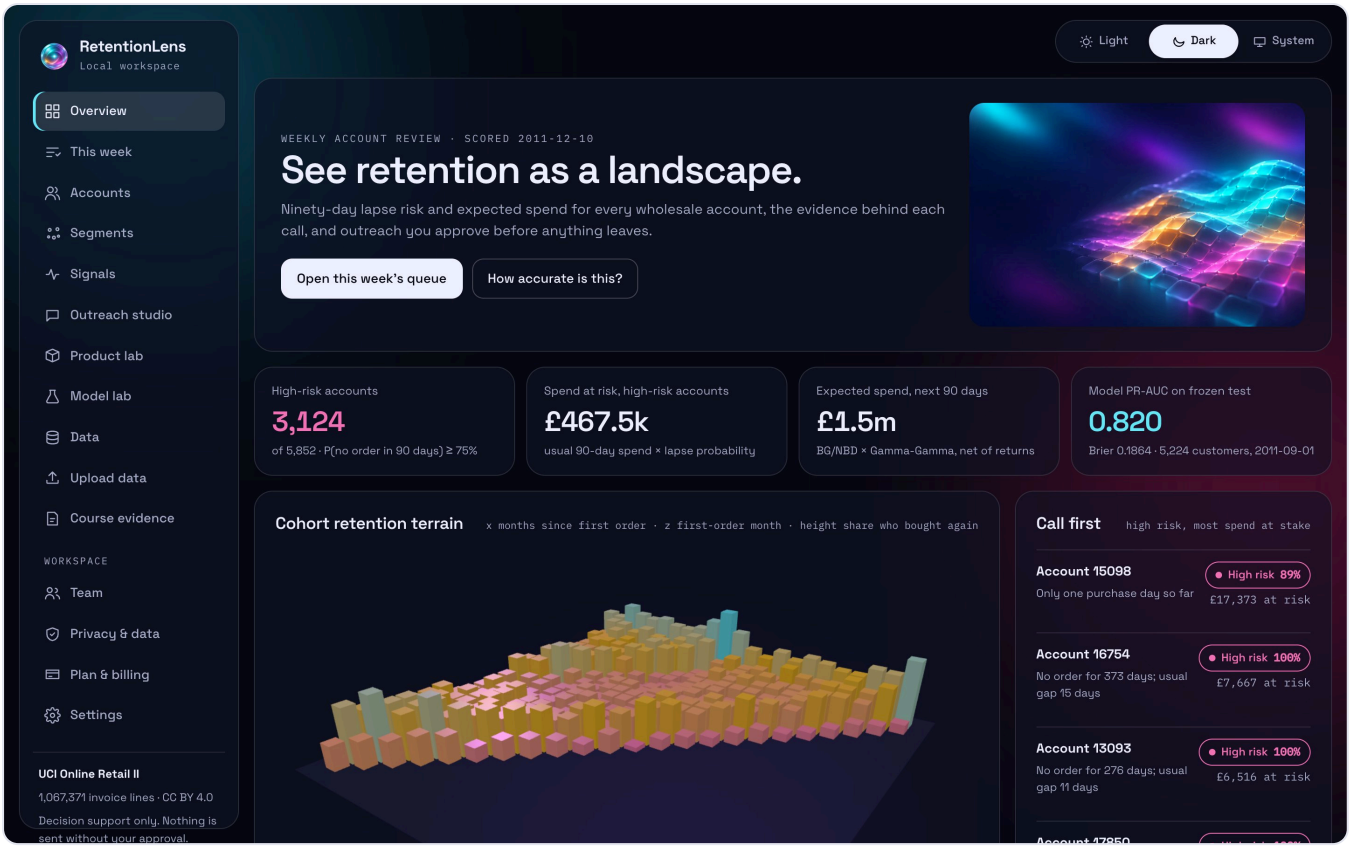


Figure 1. Overview screen (dark theme): key figures and the cohort retention terrain.

2 · PROBLEM AND USER

Who it is for and why it matters

The target user is the owner or account manager of a small wholesaler with hundreds of trade accounts and no data team. Between 65% and 67% of known accounts placed no order in the following 90 days at the development cutoffs, and lapses are usually noticed only after revenue has gone.

Desk research found a gap between self-serve direct-to-consumer tools (which need e-commerce events and hundreds of buyers) and distributor platforms (quote-only, ERP-integrated). No tool combined CSV-first input, small-sample uncertainty, approval-gated AI drafts and published pricing. Buyer, workflow and price are **hypotheses**: no interviews or pilot have been run yet.

3 · DATA

One licensed source, every problem counted

MEASURE	VALUE
Invoice lines	1,067,371
Date range	2009-12-01 to 2011-12-09
Invoices · customers · countries	53,628 · 5,942 · 43
No customer ID	243,007 (22.8%)
Cancellations · negative quantities	19,494 · 22,950
Duplicates (across sheets / within a sheet)	22,202 / 12,133
Rows usable for customer models	794,163 (74.4%), 5,875 customers

The two workbook sheets overlap on 1–9 December 2010, so the published row count includes a duplicated week. Rows are flagged, never silently dropped. The archive is SHA-256 checked (`572e36277c2390fb…`).

Upload your own invoices, review a short queue, measure the effect

A wholesaler uploads a CSV or Excel export with one row per invoice line. The wizard guesses the column mapping (editable), then a check report counts rows read and rejected, cancellations, duplicates and date-format assumptions before anything is scored. Missing customer fields are never filled with defaults. Scoring runs in a background worker with the same seasonal model and value ranges as the reference data.

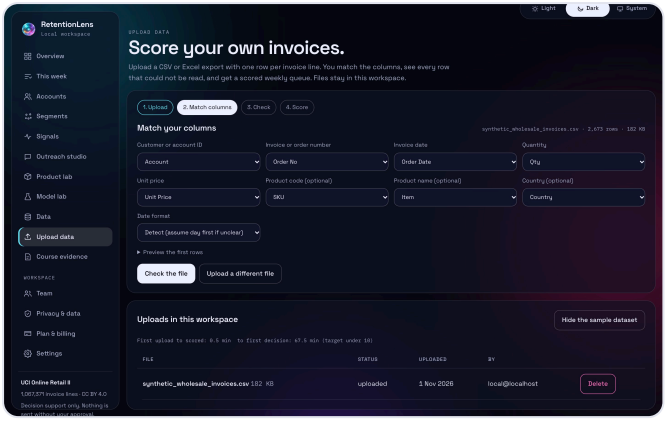


Figure 5. Upload step 2: match columns (dark). SYNTHETIC sample file.

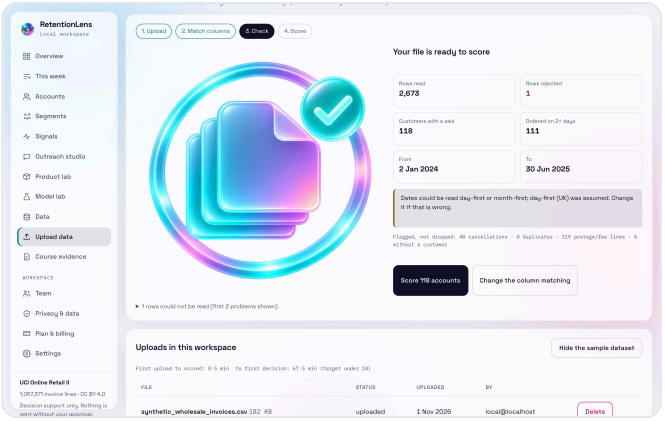


Figure 6. Upload step 3: the check report before scoring (light). SYNTHETIC sample file.

Each week the queue lists the accounts with the most spend at risk. A random share of eligible accounts (setting `RL_HOLDOUT_SHARE`, default 20%) is held out by a stable hash of workspace, account and week and is not shown. After the observation period the reorder rates of contacted and held-out accounts are compared with a 95% interval. **No uplift result exists yet**: it needs several weekly queues and later uploads from a real customer.

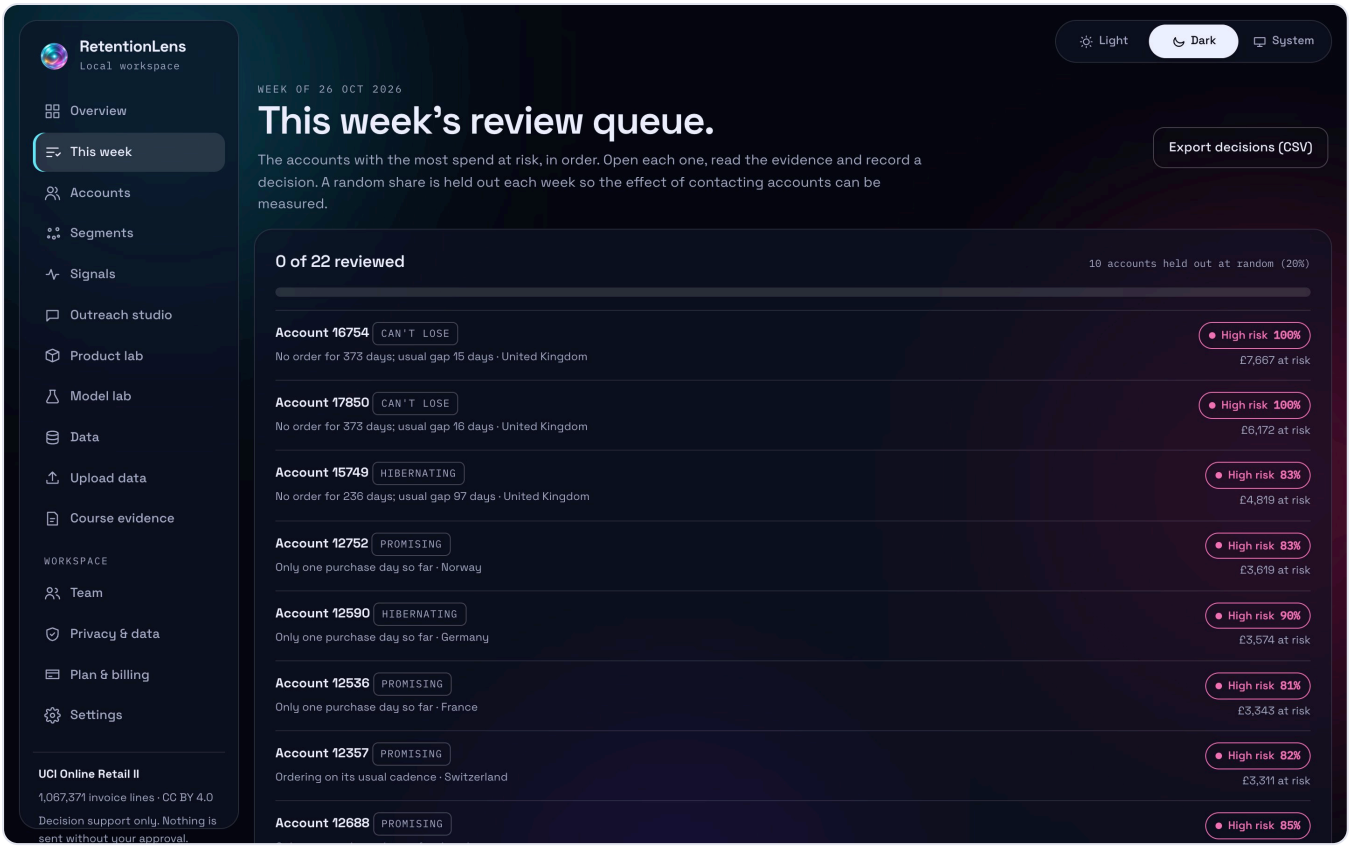


Figure 7. This week's review queue on the reference data (dark).

The screenshots use apps/web/public/samples/synthetic_wholesale_invoices.csv, which is SYNTHETIC. The upload was deleted after capture.

4 · METHOD

A protocol fixed before any result

- Snapshots at 2010-09-01, 2010-12-01, 2011-03-01, 2011-06-01, 2011-09-01; development cutoffs 2010-12-01, 2011-03-01, 2011-06-01; **frozen test 2011-09-01, scored once**.
- Features use only data before the cutoff and labels the next 90 days (a unit test proves no leakage). Lapse = no purchase day in the window; value = net spend, floored at £0.
- Models: recency rule, logistic RFM, BG/NBD + Gamma-Gamma (PyMC-Marketing 1.3, MAP), LightGBM 4.7. Fixed settings, no tuning.
- A model "wins" only if better on at least 3 of 4 cutoffs.

5 · RESULTS

Lapse prediction (PR-AUC)

MODEL	2010-12-01	2011-03-01	2011-06-01	2011-09-01 (TEST)
Recency rule	0.804	0.797	0.847	0.762
Logistic RFM	0.867	0.848	0.886	0.814
BG/NBD	0.865	0.848	0.889	0.820
LightGBM	0.836	0.826	0.882	0.819

Calibration (Brier score, lower is better)

MODEL	2010-12-01	2011-03-01	2011-06-01	2011-09-01
Recency rule	0.2777	0.1992	0.1777	0.2131
Logistic RFM	0.2522	0.1822	0.1642	0.1875
BG/NBD	0.1848	0.1721	0.1535	0.1864
LightGBM	0.2693	0.1826	0.1555	0.1938

90-day spend ranking (normalised Gini)

MODEL	2010-12-01	2011-03-01	2011-06-01	2011-09-01
Last 90 days	0.768	0.682	0.758	0.717
BG/NBD + Gamma-Gamma	0.802	0.811	0.851	0.825
LightGBM	0.807	0.775	0.845	0.807

> The test window is the Christmas peak. All P2 models over-predict lapse there and under-predict spend. Section 5b reports the seasonal correction added afterwards (R5), selected on development cutoffs only.

Fixing the peak-season gap, honestly

Experiment run 2026-11-01, seed 42. Models were chosen on the development cutoffs 2010-12-01, 2011-03-01, 2011-06-01 only; the 2011-09-01 window had already been scored once in P2 and appears here only as a labelled post-hoc look.

32.9% → 16.8%
mean total-spend bias, dev cutoffs

74.2% → 80.0%
80% range coverage on the peak window (post hoc)

0 / 3
dev wins for ZILN on lapse PR-AUC

79%
peak spend gap closed, post hoc (optimistic)

CUTOFF	MODEL	PR-AUC	BRIER	VALUE MAE (£)	TOTAL-SPEND BIAS
2010-12-01	BG/NBD	0.865	0.1848	366.0	+61.8%
2010-12-01	BG/NBD + seasonal	0.865	0.1736	323.4	+30.4%
2010-12-01	ZILN network	0.862	0.2480	558.0	+153.7%
2011-03-01	BG/NBD	0.848	0.1721	289.8	+28.5%
2011-03-01	BG/NBD + seasonal	0.848	0.1701	279.8	+19.7%
2011-03-01	ZILN network	0.812	0.1873	329.1	+37.3%
2011-06-01	BG/NBD	0.889	0.1535	270.5	+8.5%
2011-06-01	BG/NBD + seasonal	0.889	0.1526	265.4	+0.2%
2011-06-01	ZILN network	0.884	0.1554	295.4	+22.1%
2011-09-01 (post hoc)	BG/NBD	0.820	0.1864	359.7	-33.5%
2011-09-01 (post hoc)	BG/NBD + seasonal	0.820	0.1785	360.0	-7.1%
2011-09-01 (post hoc)	ZILN network	0.821	0.1862	372.8	-28.7%

- **Adopted:** a seasonal multiplier on BG/NBD + Gamma-Gamma (better Brier and value MAE on 3 of 3 dev cutoffs; ranking metrics cannot change because every account is rescaled equally).
- **Adopted (R5b):** 80% split-conformal value ranges calibrated on residuals pooled from the previous 90 days and the same window last year. Dev coverage 79.5%, 85.1%, 84.7%.
- **Not adopted:** the ZILN network (value MAE wins 0 of 3); kept as the documented deep-learning challenger. Seasonal features for LightGBM were mixed and also not adopted.

> **Disclosure.** Calibrating on the previous 90 days alone was best on the development cutoffs but covered only 74.2% of accounts on the peak window; the pooled version covered 80.0%. The peak window had already been seen, so choosing the pooled version was informed by it. The ZILN network went through three versions after dev results were seen; all three are listed in reports/r5_results.md.

Who to call, who to nurture

SEGMENT	ACCOUNTS	MEDIAN NET SPEND	EXPECTED 90-DAY SPEND	HIGH RISK	SUGGESTED PLAY
Champions	1,239	£3,892	£1,000,902	0%	Thank them, offer early access to new ranges, ask for referrals.
Loyal	735	£1,378	£155,832	6%	Keep cadence; suggest complementary lines from basket rules.
Needs attention	631	£897	£112,284	40%	Review the evidence before deciding.
Promising	670	£504	£97,510	67%	Second-order nudge with lines they have not tried.
Hibernating	1,435	£281	£53,126	100%	Low-cost reminder only; do not spend sales time.
New	236	£244	£40,484	61%	Onboarding follow-up after the first delivery.
At risk	689	£573	£32,282	95%	Check-in with a reason to reorder (new season, stock held).
Can’t lose	217	£2,041	£22,648	64%	Personal call from the account manager this week.

k-means on log recency, frequency and spend chose k = 4 by silhouette score. The lookalike model scores customers whose first order was in the last 90 days for the chance of becoming a top-20% first-year spender: test AUC 0.821 on 1,285 later customers (time-based: earliest 70% of first-order dates train, latest 30% test).

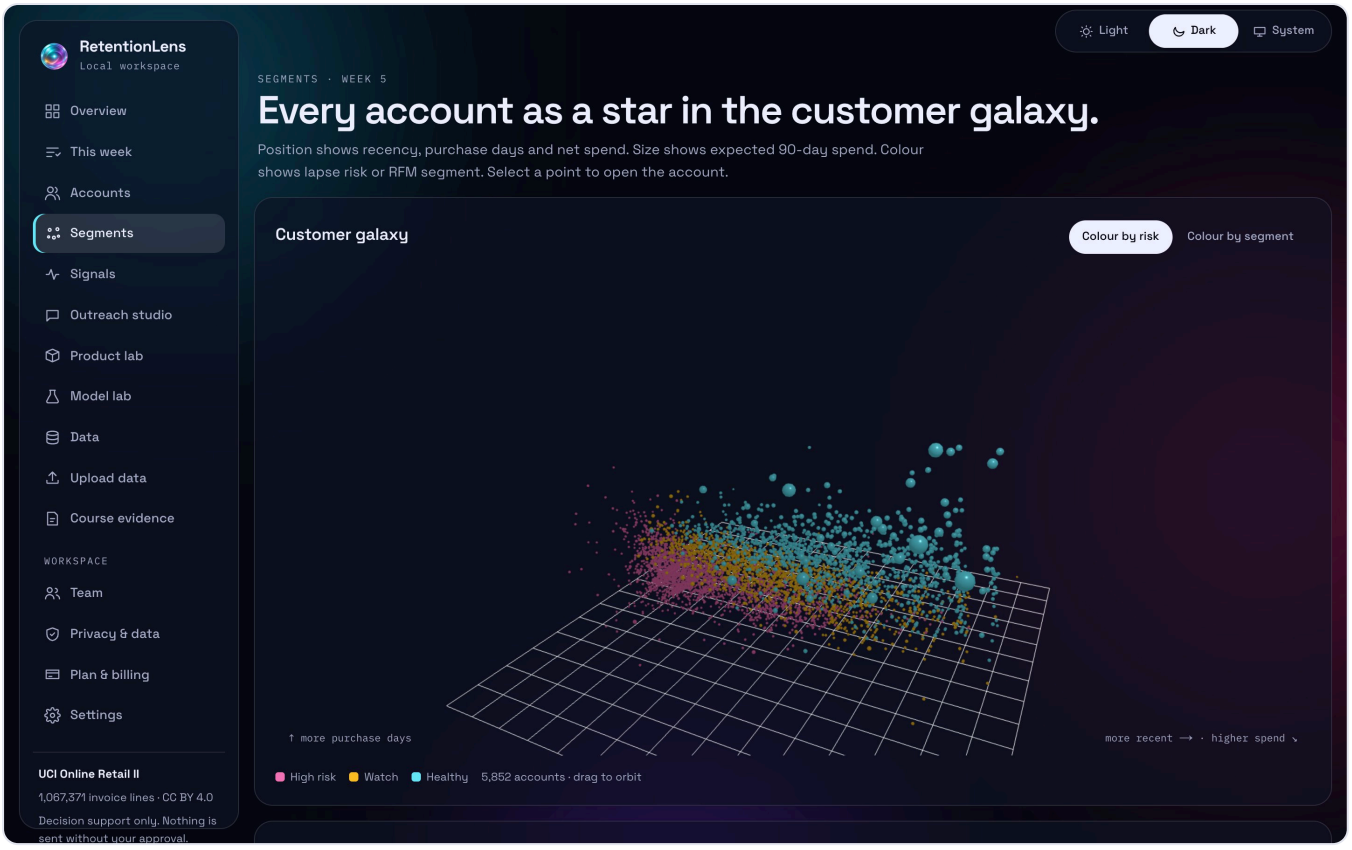


Figure 2. The 3D customer galaxy: every account positioned by recency, purchase days and spend, coloured by lapse risk.

Signals around the invoices

THEME	SAME-MONTH R	R WITH INTEREST 1 MONTH LATER	SALES PEAK	INTEREST PEAK
Christmas	0.34	0.90	11	12
Candles and lights	0.63	0.78	11	12
Bags	0.28	0.23	11	8
Mugs and cups	0.48	0.43	11	11

Wikipedia pageviews (CC0) cover 2016–2025 while invoices cover 2009–2011, so only the seasonal shape is compared. The co-purchase network links the top 120 products into 3 communities (modularity 0.183); it shows products bought together, not social influence.

Social data is synthetic. Licensed review and social sources (Yelp, Reddit, Amazon reviews) cannot be published in a public repository, so 600 posts were generated from real product names and labelled SYNTHETIC everywhere; 5 of 5 prompt-injection posts were flagged. A polite scraper (robots.txt, identifying user agent, delays, caching) collected 60 listings from the toscrape.com practice site.

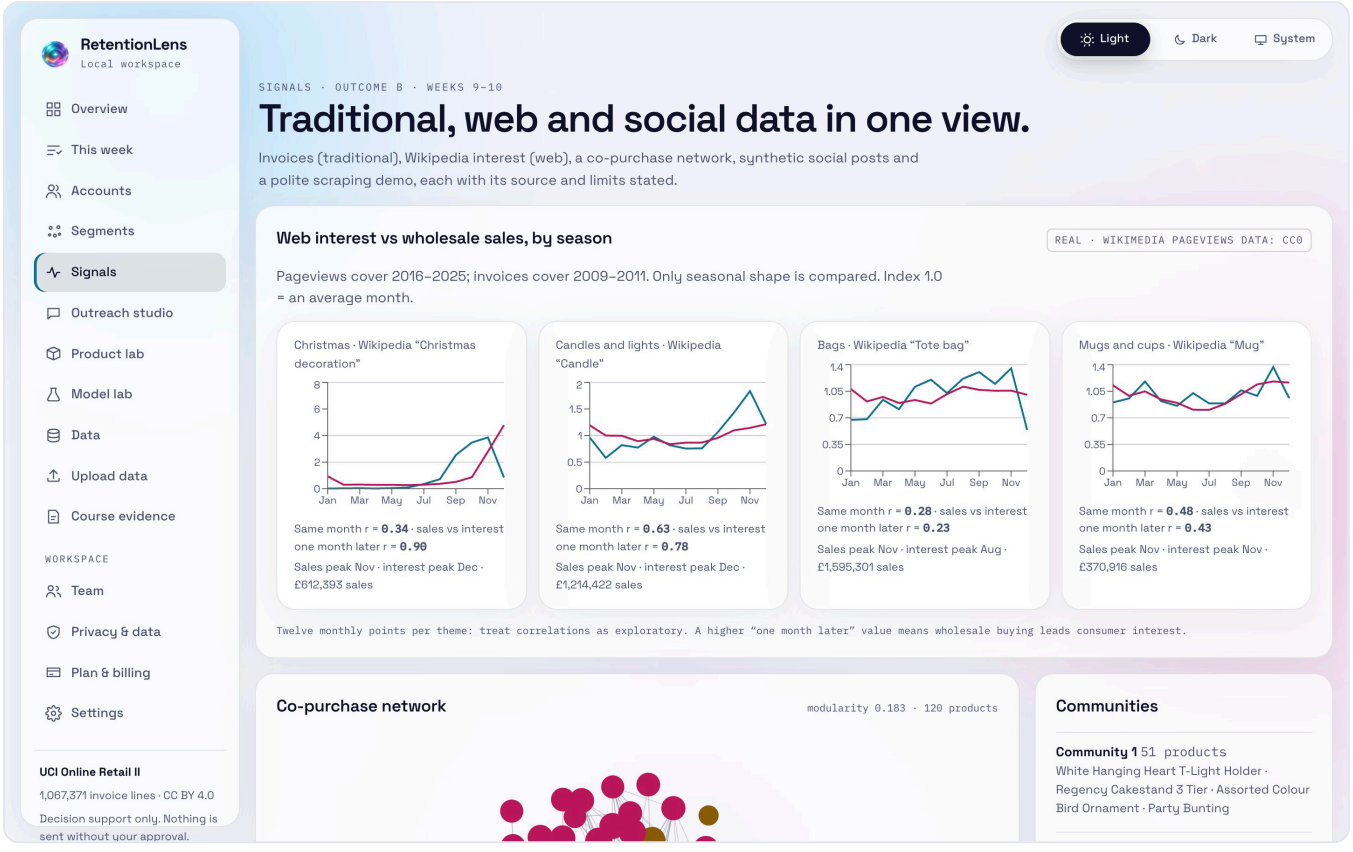


Figure 3. Signals page (light theme): wholesale sales against Wikipedia interest by theme.

Grounded drafts, human approval

The analytics core supplies every fact; the local model (qwen2.5:7b via Ollama) only writes the words. No free text from customers or posts enters the prompt. Each draft passes a deterministic checker and a second-pass review, then needs a person to approve it with a UK PECR consent basis. Approvals are idempotent, every action is audit-logged, and there is no send tool: approved drafts are exported as CSV.

CHECK (LIVE RUN, 16 DRAFTS)	PASSED
Numbers match the model's facts	15/16
No pressure or false urgency	16/16
No discounts or offers	16/16
No personal contact data	16/16
Uses name placeholders	16/16
Length 40–180 words	16/16
Second-pass review (tone, manipulation)	1/16

The second-pass reviewer is strict. Whether its flags are right is unknown until a person labels the drafts; the labelling sheet and scoring script (Cohen's kappa) are in the repository.

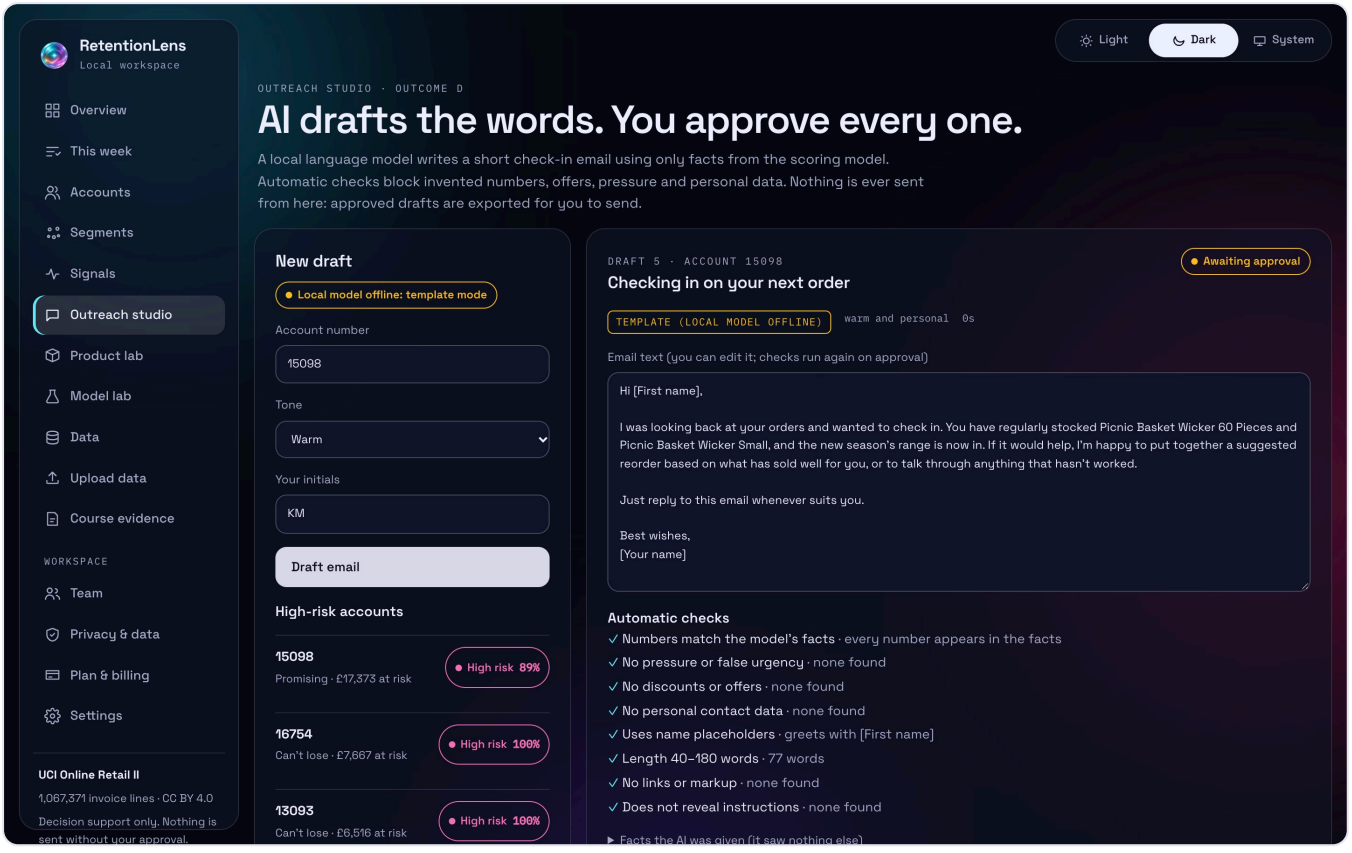


Figure 4. Outreach studio: a local-model draft with its automatic checks and the approval form.

9 · RESPONSIBLE AI AND ACCESSIBILITY (CLO4)

Built so it cannot email a customer by itself

- **Prompt injection (OWASP LLM01):** no free text in prompts; injection posts detected and tested.
- **Excessive agency:** no send tool; export only after approval.
- **Consent:** companies under legitimate interest; sole traders need consent or the soft opt-in, otherwise approval is blocked.
- **Honesty:** synthetic data labelled; every chart has a source line; limitations stated in the app.
- **Accessibility:** axe-core WCAG 2.2 AA scan reports 0 violations on 9 pages in both themes; layout verified at 320–768 px.

10 · PRODUCT LAB (WEEKS 7–8)

What sells together

IF THEY BUY	THEY ALSO BUY	CONFIDENCE	LIFT
Green Regency Teacup And Saucer	Pink Regency Teacup And Saucer	67%	28.4×
Pink Regency Teacup And Saucer	Green Regency Teacup And Saucer	84%	28.4×
Bathroom Metal Sign	Toilet Metal Sign	60%	26.2×
Toilet Metal Sign	Bathroom Metal Sign	76%	26.2×
Pink Regency Teacup And Saucer	Roses Regency Teacup And Saucer	80%	23.9×
Roses Regency Teacup And Saucer	Pink Regency Teacup And Saucer	56%	23.9×

Price elasticities from weekly log-log regressions come out implausibly large (around −15 to −20) because wholesale price breaks give big orders lower unit prices. They are presented as a lesson in confounding, not as pricing guidance.

Built and tested locally; not yet hosted

AREA	WHAT EXISTS	EVIDENCE
Workspaces and roles	One workspace per company; owner, reviewer and viewer roles; every query scoped to the workspace; sign-in throttling and weak-password refusal	66 API tests collected
Row-level security	Supabase migration with Postgres policies; a two-tenant test raises on any cross-tenant read or viewer write	CI job `rls` applies the migration to a local Supabase stack
Load test	20 concurrent users for 30 s: 10,261 requests, 342 per second, 0 errors	p95 270.9 ms vs 500 ms target (met); reports/load_test.json
Restore drill	Backup restored; checksum and integrity check passed; row counts matched in all 14 tables	SYNTHETIC seeded database; reports/restore_drill.md
LLM red team	OWASP Top 10 for LLM Applications (2025) mapped to deterministic controls	23 cases in CI; docs/security/LLM_RED_TEAM.md
Browser tests	Every route, both themes, accessibility and mobile layout; one live upload → score → review → draft → approve flow	52 static + 1 live Playwright tests

Test counts were collected on 2026-11-01 with the commands in design/test_counts.json (61 ML, 66 API, 52 + 1 browser). The load test ran on a local machine and the restore drill on a small synthetic database; neither has run on hosted infrastructure because no Render, Supabase or Stripe account exists yet.

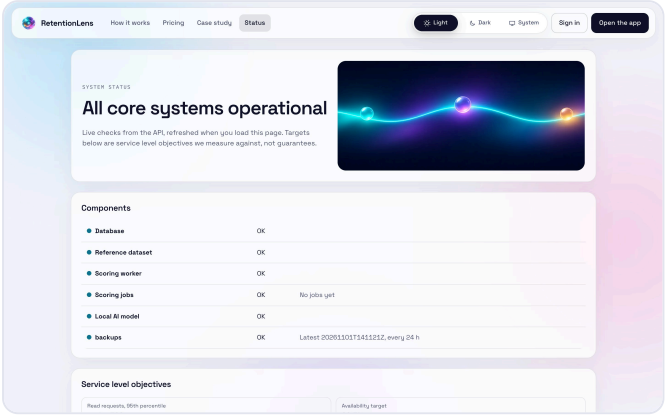


Figure 8. Status page: live component checks and service-level targets (light).

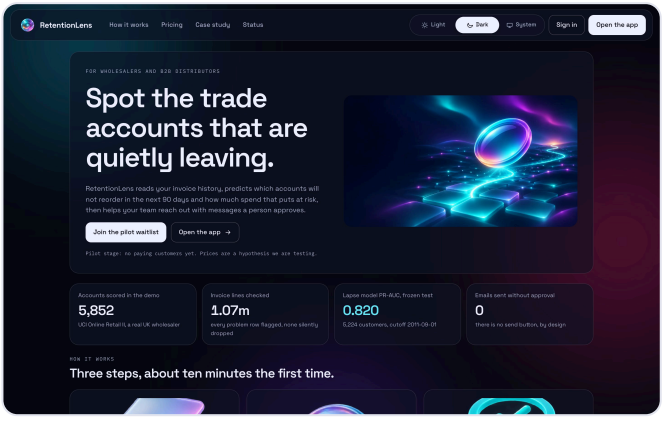


Figure 9. Public landing page (dark). Pilot stage: no paying customers.

11 · COURSE OUTCOME MAPPING

Where each outcome is shown

OUTCOME	EVIDENCE IN RETENTIONLENS
A · Marketing research and predictive analytics	Data profile and card; rolling-cutoff lapse models; cohort terrain
B · Traditional, web and social data	Wikipedia seasonality; co-purchase network; synthetic social posts; scraper
C · CLV, churn, retention, new customers	BG/NBD + Gamma-Gamma; RFM segments; k-means; lookalike scoring
D · Generative AI for marketing decisions	Outreach studio with checker, critic, consent gate and live evaluation
Weeks 1–11	Mapped topic by topic on the Course evidence page
Labs (weeks 2, 4, 6, 8, 10)	Scaffolds and generated EVIDENCE.md in labs/ · **[OWNER: lab reports to be written]**

12 · WHAT IS STILL MISSING

The software exists; the market evidence does not yet

- **No customer interviews**: buyer, workflow and price remain hypotheses (kit ready in docs/research).
- **No usability sessions**: the ten-minute first-run target is untested with real users.
- **No pilot and no customers**: the holdout queue has produced no uplift result, so there is no causal evidence that outreach helps.
- **No hosting**: Render, Supabase and Stripe test accounts are not created; everything ran locally or as a static demo.
- The AI reviewer is uncalibrated (no human labels); social data is synthetic; one UK giftware wholesaler, 2009–2011.

Reflection

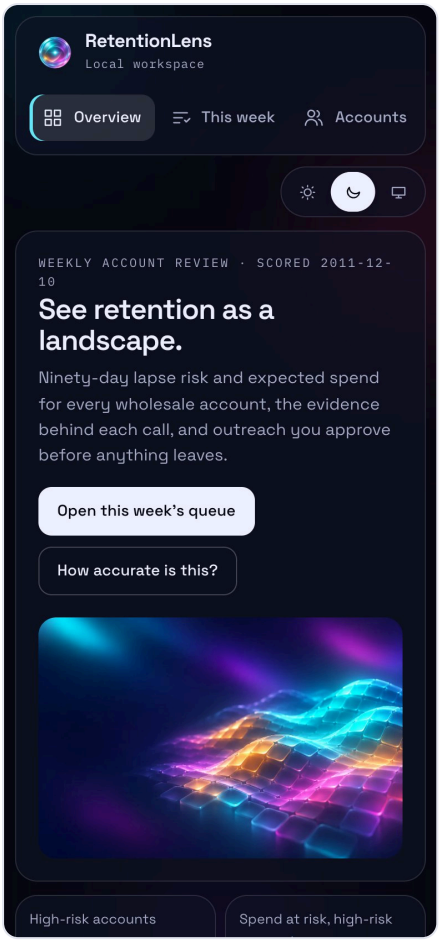
> **[OWNER: reflection and learning narrative to be written by the student. Not generated.]**

References

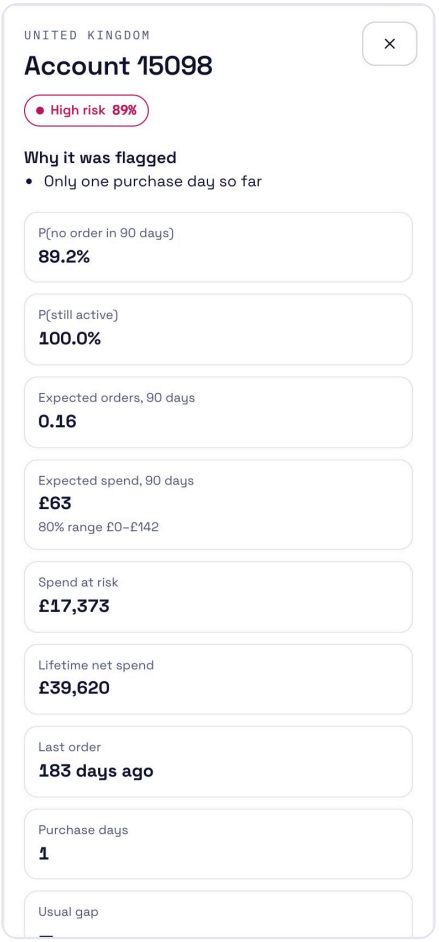
Chen, D. (2019). Online Retail II. UCI Machine Learning Repository. doi:10.24432/C5CG6D · Fader, Hardie & Lee (2005). Counting your customers the easy way. Marketing Science 24(2) · Wang, Liu & Miao (2019). A deep probabilistic model for CLV prediction. arXiv:1912.07753 · Hwang (2019). Hands-on Data Science for Marketing. Packt · Winston (2014). Marketing Analytics. Wiley · Grigsby (2018). Marketing Analytics. Kogan Page · OWASP Top 10 for LLM Applications (2025).

APPENDIX · MOBILE

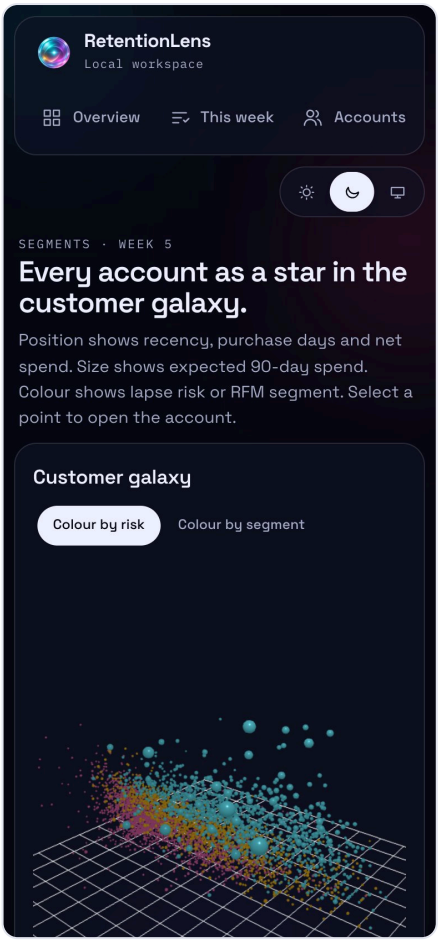
The same product on a phone



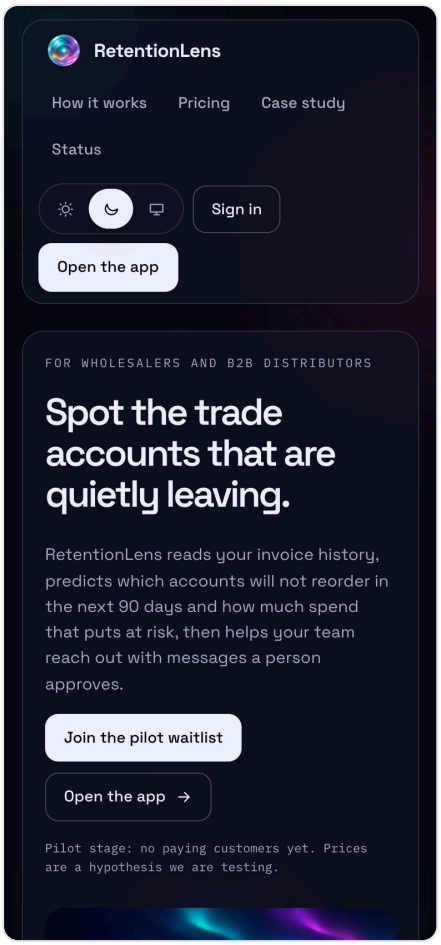
Overview, 390 px, dark



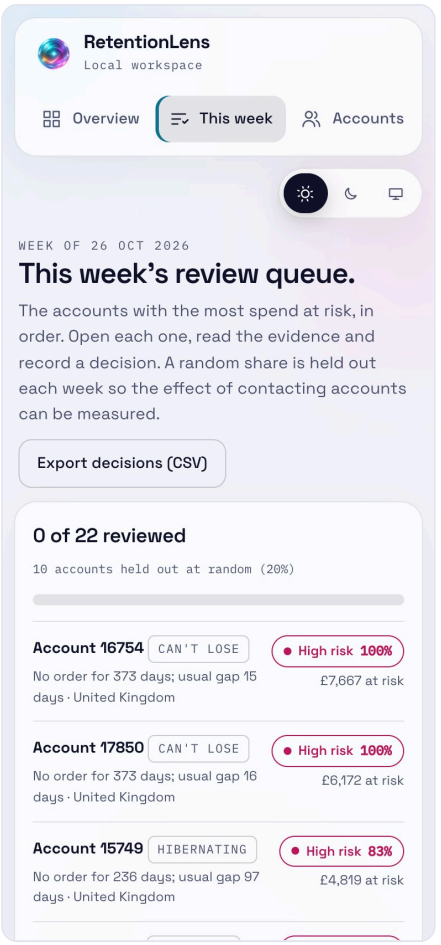
Account evidence, 390 px, light



Segments, 390 px, dark



Landing page, 390 px, dark



Weekly queue, 390 px, light

Illustrations generated by the author with ChatGPT Images (gpt-image-2). Screenshots captured from the running application on 2026-11-01.